

DRAFT Memo

To: Mr. Philip Paradis, Jr., PE, LEED AP, CPSWQ
BETA Group, Inc.

From: Lee Wooten, P.E.

Date: December 22, 2023

Re: Dam Design Review Assistance for 42 Mill St. Notice of Intent, Westwood, MA

The purpose of our memo is to assist BETA Group, Inc. (BETA) and the Town of Westwood with the review of a Notice of Intent (NOI) for a proposed development at 42 Mill Street in Westwood, MA, relative to dam safety issues.

Background

The Town of Westwood (The Town) has received a NOI for site changes at 42 Mill Street. The site changes include construction of a concrete spillway repair structure, a riprap-lined auxiliary spillway structure, and repair of an existing small embankment. The Town engaged BETA to review the NOI and advise them concerning the technical aspects of the NOI.

Our review focused on the information presented about the dam design, which was summarized in the Goddard Consulting letter of September 8, 2023, Re: Peer Review Comment Response, 42 Mill Street, Westwood, MA (Ref. 1). The information in Ref. 1 provided some descriptions of the proposed construction including:

- Repair of the concrete spillway next to the left¹ abutment, which includes rebuilding a 6-foot section of the 14-foot-wide spillway.
- A rebuilt auxiliary spillway ("Overflow spillway") next to the right abutment, which would have a riprap lining.
- Repair of the existing dam embankment, which is described as being 4.6 ft high. The repairs would include removal of trees (but not large roots), backfilling of holes and depressions with clean gravel borrow, covering the embankment with clean loam. The embankment section in Ref. 1 shows that a breached area by the concrete spillway would be covered with 4 inches of

¹ The terms left and right are given relative to the perspective of an observer who is looking downstream.

topsoil over 8 inches of clean granular fill over low permeability fill as needed with a downstream slope of a 2 horizontal to 1 vertical (2H:1V) grade and a 3-foot-wide crest at El. 192±.

GEI Tasks

- Site visit – We visited the site on December 19, 2023, to view the existing conditions. We viewed the dam and pond from the right (west) abutment and from the downstream area along Millbrook Road. We could not view the existing concrete spillway because of the stream flows and the private property on the left abutment.
- Review and Evaluation of Ref. 1 – We reviewed Ref. 1 to gain a better understanding of the dam design and proposed construction. We also evaluated the design relative to dam safety standards and good engineering practice. Our evaluation comments can be found in the next section.
- Memo preparation – We prepared this memo to describe our review and evaluation.
- Meetings – If needed, we will attend meetings of the Conservation Commission.

Evaluations / Recommendations

We provide the following evaluations and recommendations about the proposed changes to the dam relative to dam safety and good engineering practice. Our first set of comments describe deficiencies in the design of the dam relative to the state-of-the-practice for dam safety.

- The drawings do not adequately describe the proposed changes as noted below:
 - There is no plan that shows how the new concrete spillway repair is located relative to the existing concrete spillway and the embankment, and the section and elevation do not show how the two spillway structures will be connected or abutted.
 - There is no cross-section for the overflow spillway along the direction of flow.
 - The final grades of the embankment are not shown on the plan.
 - No specifications are provided for most of the materials, especially earthwork materials.

- The new concrete spillway design lacks the following elements:
 - A foundation design. There is no subsurface information for design of the structure foundation, and there is no description of how the contractor should dewater, evaluate, protect, and prepare the foundation. Also, the proposed foundation will be susceptible to frost heave because of the shallow depth of the foundation.
 - Tie-ins to the adjacent embankments. Seepage control should be an element of this design.
 - Stability design for normal pool, flood, and ice loads on the structure.
 - Stress analysis for the new structure, which has a modest (10-inch) thickness, for various loading conditions.
 - Filtered seepage control elements. Without filtered seepage control drains or drainage layers, the spillway will be susceptible to internal erosion undermining.
 - Erosion protection in the downstream plunge pool and channel. Without erosion protection in the downstream plunge pool, the spillway will be susceptible to undermining.
- The new overflow spillway design lacks the following elements:
 - Erosion protection design. The design calls for riprap lining of the spillway but does not specify the size of the riprap or provide an analysis that the riprap will be stable for the anticipated spillway flows.
 - Riprap bedding. The design lacks a layer that would prevent underlying soil from eroding through the riprap.
 - Filtered seepage control elements. Without filtered seepage control drainage layers, the overflow spillway will be susceptible to internal erosion undermining.
 - Erosion protection in the downstream plunge pool and channel. Without erosion protection in the downstream plunge pool, the overflow spillway will be susceptible to undermining.
 - As noted previously, there is no cross-section of the overflow spillway in the direction of flow. Details about the design elements along such a section are critical to the performance of the overflow spillway.
 - Access for maintenance. Access for maintenance will have to be across the overflow spillway. The proposed 1H:1V slopes for the sides of the overflow spillway will make access difficult for grass

mowing equipment and large landscape/earth moving equipment that will be need for maintenance of the embankment and removal of future beaver dams.

- The repaired embankment design lacks the following elements:
 - Freeboard during the design flood. The submitted design indicates that the pond will rise to about the crest of the repaired embankment (El. 192) for the anticipated 100-year flood event. Larger floods or wind-driven waves would result in overtopping of the embankment. We found that a slightly larger 100-year flood (201 cfs instead of 172 cfs) is predicted for the site using the USGS StreamsStats model. Beaver damming upstream of the spillways is likely again and would increase the overtopping potential.
 - Design for seepage and stability. No design is presented to demonstrate that the embankment will be stable for various pond levels and seepage conditions. No information is presented about the existing embankment materials that will remain, an important element for the embankment seepage and stability. No specifications are provided for the new materials that will be used to construct the modifications.
 - Removal of large roots. The design indicates that the trees on the dam will be cut and that the roots will be left in place. These roots will eventually rot with the potential to create seepage paths that could result in internal erosion through the embankment.
 - Appropriate soil materials in the embankment repair section. The embankment repair design section shows that an 8-inch-thick layer of clean granular fill will be placed across the dam from upstream to downstream. During high pool levels, this layer will more readily convey seepage through the dam with the potential to create instability or internal erosion on the downstream slope.
 - Obviously stable embankment slopes. The design slopes of 2H:1V are relatively steep for dam embankments and could be susceptible to instability under various seepage conditions.

Relative to overall dam safety and liability, we have the following comments.

- The Town should consider its liability in any decision allowing construction of a dam. Ref. 1 correctly notes that at a height of 4.6 ft, the dam will not cross the 6 ft criterion for regulation under the Massachusetts Dam Safety Regulations (302 CMR 10). However, under large flood events, a large

breach of the dam could result in severe flooding across the downstream Millbrook Road, a private drive crossing, and High Street. High Street is secondary highway, and its presence would mandate that the dam be considered as having a Significant Hazard potential if the dam were regulated. Failure of a dam that is constructed with the Town's review could make the Town liable for costs associated property damage or personal injury.

- The existing dam, which was constructed by men and beavers without review and approval by the Town, is in unsafe condition. The existing trees could uproot and breach the dam. The existing roots provide some erosion protection that might limit erosion and the size of a breach if one occurs, but those roots also degrade the dam by displacing earth and eventually rotting. Beaver activity has raised the pond well above the existing spillway weir, increasing the potential for overtopping.

The Town should weigh these factors in reaching their decision about the proposed dam project. Removal of the existing dam without replacement would be the safest option. If the Town allows dam construction to proceed, we recommend that the design of the dam modifications meet the Massachusetts Dam Safety regulations for regulated dams and address the comments that we listed. The approval should also mandate that the dam be maintained and that beaver damming upstream of the spillways be prevented.

References:

1. Goddard Consulting (2023). Letter Re: Peer Review Comment Response, 42 Mill Street, Westwood, MA. September 8.
2. Massachusetts Department of Conservation and Recreation (2017). 302 Commonwealth of Massachusetts Regulations 10.00 (Dam Safety). February 10.